

Dermatologists show highest melanoma diagnostic performance with AI support

Written by [Peter Morales-Brown](#) on April 9, 2026 — Fact checked by [Jill Seladi-Schulman, Ph.D.](#)



Evidence suggests that AI tools assisting dermatologists can improve diagnostic performance for skin cancer. Image credit: PER Images/Stocksy

- **A study suggests that artificial intelligence (AI) showed comparable performance to dermatologists in detecting melanoma, with similar sensitivity and specificity in prospective clinical studies.**
- **However, AI demonstrated a higher specificity in direct comparisons, suggesting it may help reduce unnecessary biopsies by better identifying benign lesions.**
- **The combination of dermatologists using dermoscopy with AI support showed the highest diagnostic performance.**
- **Despite the promising results, current evidence remains limited and potentially biased, highlighting the need for larger, real-world studies before widely adopting AI in clinical practice.**

Experts often consider [melanoma](#) to be the [most serious](#) type of [skin cancer](#) due to the likelihood of [metastasizing](#). Evidence suggests that roughly [112,000 Americans](#) receive a diagnosis of melanoma each year, and more than [1.5 million people](#) in the U.S. are living with melanoma.

While only accounting for about 1% of skin cancers, melanoma causes a large [majority](#) of skin cancer deaths.

Early detection of melanoma can be critical, as the 5-year survival rate for early melanoma is [94%](#). A common approach for detection is via a dermatologist's diagnosis. However, reliance on specialists can make it difficult for people to receive a timely diagnosis.

Tools such as a [dermatoscope](#) can greatly [improve](#) the accuracy of skin cancer detection among dermatology clinicians. However, could AI further improve detection rates?

A study, published in [JAMA Dermatology](#) , suggests that AI may soon play a larger role in melanoma detection. However, the research team cautions that more rigorous testing is still necessary before widespread adoption in routine care.

ADVERTISEMENT

AI performs on par with dermatologists

In recent years, there has been growing interest in AI-based analysis as a [diagnostic aid](#) for melanoma. However, the use of AI in real-world clinical settings remains controversial. While studies suggest that AI could be utilized in these settings, real-world evidence has been limited.

To address this, researchers conducted a systematic review and meta-analysis of 11 prospective studies. These studies involved more than 2,500 participants and 50 dermatologists.

The findings suggest that for melanoma diagnosis, dermatologists achieved a sensitivity of 78.6% and specificity of 75.2%. AI systems alone reached 80.9% sensitivity and 75.6% specificity.

These results suggest that AI demonstrated a diagnostic performance comparable to that of dermatologists.

Notably, in one study, dermatologists assisted by AI achieved 91.9% sensitivity and 83.7% specificity. This indicates that AI could serve as a valuable tool to assist, rather than replace, clinicians.

[Tanya Evans](#), MD, board certified dermatologist and medical director of the Skin Cancer Program at the Melanoma Clinic at MemorialCare

Saddleback Medical Center in Laguna Hills, CA, who was not involved in the study, notes the key takeaway is not that AI could replace dermatologists, but can reliably function as a clinical adjunct:

“The most important implication is that diagnostic accuracy can be meaningfully improved when AI is combined with clinician expertise, rather than used independently. The study emphasizes that AI is still early in validation, with bias and limited generalizability—so it is not yet ready for autonomous use.”
— Tanya Evans, MD

AI may reduce unnecessary biopsies

While comparable, the AI systems tended to show a higher specificity in head-to-head comparisons within the same clinical settings. This suggests the AI tools were better at correctly identifying benign lesions.

The higher specificity may have practical implications. When uncertain, dermatologists typically apply caution and will recommend biopsies. By contrast, AI may help to reduce unnecessary procedures by ruling out noncancerous lesions.

“This is one of the most promising and immediately actionable implications,” Evans said.

“The most immediate benefit is likely reducing unnecessary biopsies while maintaining safety. Dermatologists tend to be risk-averse (biopsy when uncertain), whereas AI is more probabilistic and specific—this complementary dynamic is where synergy happens. The most realistic near-term model is AI layered onto dermoscopy—not replacing it.”
— Tanya Evans, MD

The research team also suggest that although evidence remains limited, these findings indicate combining human expertise with AI support could offer the best results.

While the findings support the potential of AI as a decision-support tool, the researchers emphasize that the technology is still in an early validation phase. They add that larger, multicenter studies are still

necessary to determine the safety, reliability, and real-world clinical impact of these tools.



HEALTHLINE NEWSLETTER

Learn how to detect skin cancer with our free guide

There are a few changes you can watch out for on moles to help monitor for skin cancer at home. Join our melanoma newsletter for your free guide and expert guidance once a week.

Enter your email

JOIN NOW

Your [privacy](#) is important to us

Important limitations

As noted, this review used prospective studies to evaluate performance in real-time clinical settings. This means they should be more reflective of everyday practice, rather than retrospective research, which relies on preselected datasets.

This suggests that AI's performance should remain strong even outside controlled laboratory conditions. However, despite these promising findings, the study is not without its limitations.

The researchers highlight a high risk of bias, as most of the studies reviewed only included lesions already suspected of melanoma, rather than the full range a clinician will see in routine practice.

Additionally, many of the studies used simplified diagnoses, which does not reflect real clinical decision making, and differences in the study designs and datasets make comparisons challenging.

As such, these factors mean that the results may not fully translate to everyday healthcare settings.

The researchers conclude that although AI shows promise for matching dermatologist performance, more robust evidence is necessary before it becomes a standard tool in clinics. For now though, it may serve as a potential tool for clinicians to help improve early cancer detection.

ADVERTISEMENT

Find your new dermatologist

Browse local dermatologists using the FindCare tool. Select your provider based on their credentials, office locations, and accepted insurance plans.

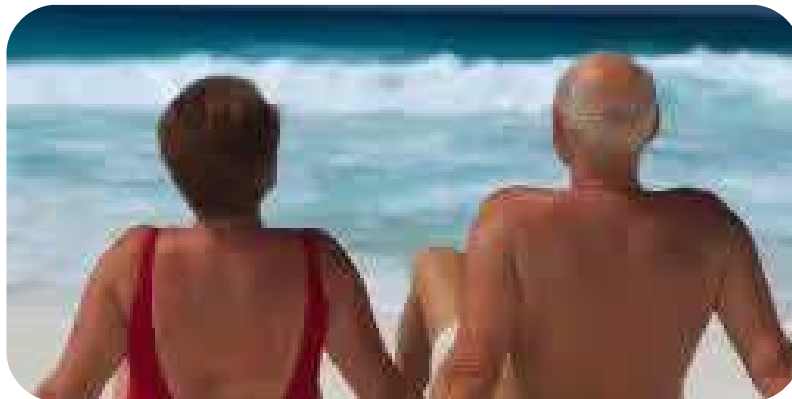
START MY SEARCH

Cancer / Oncology Medical Devices / Diagnostics Melanoma / Skin Cancer

Share this article



READ NEXT



Melanoma patients may benefit from new biomarker



Written by: [Catharine Paddock, Ph.D.](#)

A new biomarker and minimally invasive way to quickly determine whether a melanoma patient is likely to respond to a BRAF-targeted drug promises to speed up treatment decisions.



Skin Cancer: One Tumor Surprisingly More Deadly Than...



Melanoma mortality rates vary across the country



Melanoma: Immunotherapy for brain metastases...



Existing drug may improve melanoma survival



Get our newsletter

Keep up with the ever-changing world of medical science with new and emerging developments in health.

Enter your email

SUBSCRIBE

Your [privacy](#) is important to us

About Us

Health Topics

Contact Us

Health Hubs

Terms of Use

Medical Affairs

Privacy Policy

Content Integrity

[Privacy Settings](#)

Newsletters

© 2026 Healthline Media UK Ltd, London, UK. All rights reserved. MNT is the registered trade mark of Healthline Media. Healthline Media is an RVO Health Company. Any medical information published on this website is not intended as a substitute for informed medical advice and you should not take any action before consulting with a healthcare professional. [See additional information.](#)

[About](#) | [Careers](#) | [Advertise with us](#)

OUR BRANDS

Healthline
Medical News Today
Greatist
Psych Central
Bezzly